

B.3 Additional Details on the AI Algorithm

The training data is a set of tuples of images and labels. These training datasets typically rely on human input to assign the labels, which indicate whether or not a specific pattern or object is present in the image. Training is conducted through stochastic gradient descent. These algorithms build on the nested structure of the neural net to compute gradients computationally efficient via chain rule. Each training step is performed on a small batch of data so that the algorithm does not have to consider the entire dataset for each optimization step. After each round of optimization on the training set, the model performance is assessed through predictions on a hold-out validation sample. Most humans are able to recognize cars, pedestrians, and traffic light, which means that training datasets for common classification tasks are easy to come by. The same is not true for medical imaging. Classifying disease based on X-rays, CT scans, and retina scans requires the input of highly trained experts. Recently, several researchers have released large training datasets of medical images with disease labels that are extracted from written clinical descriptions (Irvin et al., 2019). The neural net has a DenseNet121 architecture. A DenseNet is a type of convolutional neural network that utilizes dense connections between layers through Dense Blocks; in these blocks, we connect all layers with matching feature-map sizes directly with each other. Images are supplied in a standardized format of 320×320 pixels. For optimization the researchers use the Adam optimizer with default β -parameters of $\beta_1 = 0.9$, $\beta_2 = 0.999$ and learning rate 1×10^{-4} . The batch size is fixed at 16 images. The training is performed for 3 epochs. The full training procedure is described in (Irvin et al., 2019).

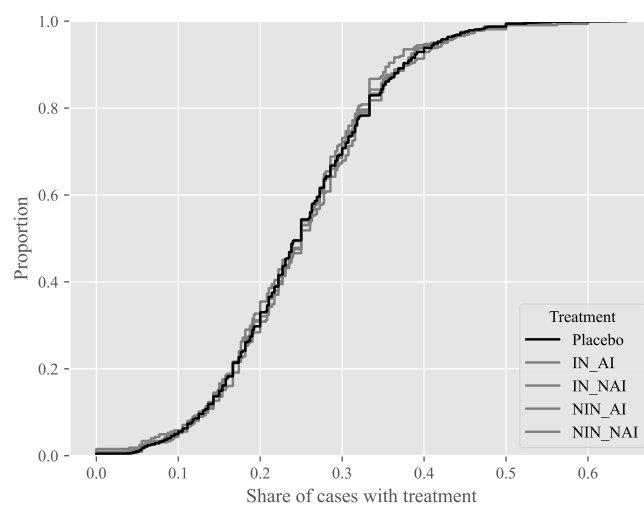
C Data Appendix

C.1 Balance Tests

We verify that the randomization occurred as expected through various balance and randomization tests. Figure C.9 plots the distribution of treatment probabilities by patient-case in Design 1. We also plot a placebo distribution that samples from the null distribution to support the claim that the randomization occurred as expected. To test this formally, we present balance tests for Design 1 and Design 2 in Table C.1 and Table C.2, respectively.³² For these balance tests, we calculate the average covariates across the four treatment arms and report p-values from the test of the joint null that the four means are equal. For Design 2, these are done within sessions as patients are balanced by design across all sessions.

³²Design 3 is balanced by design, as each radiologist reads the same cases with and without AI assistance.

Figure C.9: Distribution of Patient Treatment Probabilities in Design 1



Note: The cumulative distribution functions of patient treatment probabilities by treatment for design 1. The placebo distribution is calculated based on 100,000 draws from the null distribution. For each draw from the null distribution, we sample the number of reads the case receives from the empirical distribution and then draw the number of treatments from a binomial distribution with probability $1/4$.

Table C.1: Covariate Balance in Design 1

	Control	CH	AI	AI x CH	p-value
s_A	0.308	0.299	0.308	0.307	0.410
AIRSPACE OPACITY	0.161	0.142	0.164	0.163	0.222
CARDIOMEDIASTINAL ABNORMALITY	0.130	0.135	0.132	0.131	0.985
SUPPORT DEVICE HARDWARE	0.173	0.161	0.188	0.195	0.053
ABNORMAL	0.183	0.179	0.192	0.191	0.744
WEIGHT	185.10	186.67	185.46	184.79	0.656
TEMP	99.01	99.04	99.05	99.07	0.057
PULSE	92.11	92.68	92.28	93.02	0.012
AGE	56.50	56.15	56.31	56.70	0.882
NUMBER LABS	34.59	34.46	34.50	34.27	0.757
NUMBER FLAGGED LABS	5.863	5.914	6.060	5.982	0.646
FEMALE	0.424	0.412	0.385	0.392	0.080

Note: Balance tests of patient covariates for patients assigned to the four treatments in Design 1. Missing clinical history variables are mean-imputed. The p-values come from the joint test the mean covariates are equal across the four treatments.

Table C.2: Covariate Balance in Design 2

	Session 1	Session 2	Session 3	Session 4
s_A	0.381	0.625	0.381	0.447
AIRSPACE OPACITY	0.243	0.368	0.141	0.483
CARDIOMEDIASTINAL ABNORMALITY	0.164	0.834	0.088	0.716
SUPPORT DEVICE HARDWARE	0.760	0.770	0.714	0.794
ABNORMAL	0.265	0.624	0.722	0.330
WEIGHT	0.461	0.597	0.878	0.735
TEMP	0.107	0.245	0.437	0.654
PULSE	0.242	0.578	0.764	0.772
AGE	0.559	0.220	0.082	0.898
NUMBER LABS	0.075	0.348	0.581	0.768
NUMBER FLAGGED LABS	0.297	0.189	0.935	0.738
FEMALE	0.067	0.052	0.225	0.075

Note: Balance test p-values that the covariate means are equal across the four treatments within each session (column). Missing clinical history variables are mean-imputed.